

Decoding Customer Value

A SQL-Driven Retention Strategy

3,900

Customers

18

Variables

43%

Discount Rate

50

US States

AGENDA

01

Problem & Data Overview

Brand context · Dataset constraints

02

Loyalty Framework

2 competing definitions · Definition B wins

03

Customer Segmentation

4 value tiers · Revenue pyramid

04

Promotional Analysis

Discount dependency · Margin risk

05

Geographic Intelligence

Organic demand markets · Opportunity scores

06

Category & Behavioral Intel

Category funnel · Predictors of value

07

Power BI Dashboard Design

4-panel founder dashboard

08

Retention Playbook

Phased promo sunset · 3-phase plan

09

Ideal Customer Profile

Data-backed acquisition target

10

Strategic Recommendations

5 actions · Trade-offs stated

Why This Analysis Exists

Background

A direct-to-consumer fashion brand (clothing, accessories, footwear, outerwear) selling across the US. No physical stores. Every customer relationship managed directly. 3,900 customers, growing steadily — but customer intelligence is zero. Marketing decisions run on gut feel, not evidence.

Five Questions the Brand Cannot Answer



Who will still be buying two years from now?



Is the promo program building loyalty — or attracting bargain hunters?



Which categories drive high-frequency, high-tenure customers?



Where does the brand have organic traction vs. discount-driven volume?



What does the brand's best customer actually look like?

Problem Statement

Using only transactional and behavioral data, can the brand identify what its most valuable customers look like, measure how much of its current revenue depends on promotions, and build a data-backed retention strategy that reduces discount dependency without hurting sales?



Core Constraint

No loyalty score · No churn label · No timestamps. Every construct must be built from available variables.

Deliverables



Python

Cleaned dataset + engineered features (dependency score, value tier, satisfaction flag)



SQL

Segmentation queries answering all 5 key business questions



Power BI

Four-panel founder dashboard (pyramid, promo matrix, geo map, category funnel)



Playbook

Promo sunset plan + ideal customer profile with trade-offs



Summary

Concise executive summary (max 1 page) of findings & recommendations

Dataset & Analytical Approach

 3,900

Unique Customers

 18

Dataset Variables

 43.0%

Discount Usage Rate

 27.0%

Subscription Rate

 \$59.76

Avg. Order Value

 3.75

Avg. Review Rating

Engineered Features

frequency_score (ordinal cadence)

loyalty_score_A (behavioral)

loyalty_score_B (commercial ✓)

value_tier (quartile rank)

promo_dep_score (discount reliance)

opp_score (geo opportunity)

Data Quality Caveats

37 missing Review Ratings → imputed

Discount & Promo Code: perfectly collinear

Gender perfectly separates discount usage

Treated as structural artifact, not causal

Analytical Approach — Proxy-Based Intelligence

01

Clean & Encode

Handle 37 missing ratings · encode
categoricals · min-max normalize
spend, purchases, frequency

02

Build Metrics

Construct Loyalty A (behavioral) ·
Loyalty B (commercial) · value tier ·
promo-dependency · geo-opp

03

SQL Segments

Quartile split on Definition B → 4
equal tiers · answer all 5 business
questions via structured queries

04

Validate

Cross-check correlations · verify
segment separation · test both
definitions against spend outcomes

Two Competing Definitions — One Clear Winner

Definition A

Behavioral Loyalty — Engagement Focus

0.35 × Previous Purchases (normalized)

0.25 × Purchase Frequency (ordinal)

0.20 × Subscription Status (binary)

0.20 × Review Rating (normalized)

r = 0.004

correlation with spend



Near-zero — not actionable
for revenue decisions

Limitation: Definition A ranks frequent discount-users as Champions — misleading for margin strategy.

Definition B

✓ SELECTED

Commercial Loyalty — Value Focus

0.40 × Purchase Amount (normalized)

0.30 × Previous Purchases (normalized)

0.20 × Discount Independence (1 – discount)

0.10 × Subscription Status (binary)

r = 0.723

correlation with spend



Strong signal — directly
aligns with revenue goals

Selected: Definition B separates high-value, full-price buyers from promo-dependent low spenders.



The Value Pyramid — Who Drives Revenue?

Champion



975 customers · #1 revenue tier

\$82.91

Avg Spend

35.15

Prev. Purch.

19%

Promo Rate

16%

Sub. Rate

34.7%

Rev. Share

Loyal



973 customers · #2 revenue tier

\$65.93

Avg Spend

27.34

Prev. Purch.

34%

Promo Rate

25%

Sub. Rate

27.5%

Rev. Share

Potential



977 customers · #3 revenue tier

\$52.35

Avg Spend

23.33

Prev. Purch.

50%

Promo Rate

34%

Sub. Rate

21.9%

Rev. Share

At-Risk



975 customers · #4 revenue tier

\$37.9

Avg Spend

15.59

Prev. Purch.

68%

Promo Rate

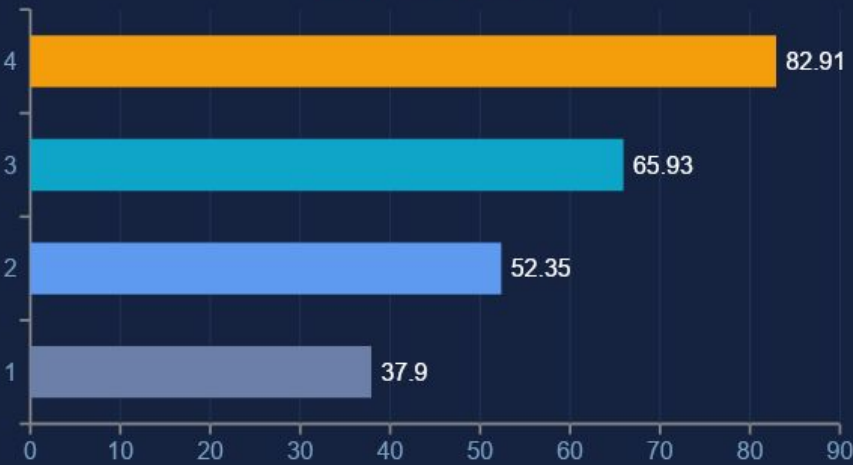
32%

Sub. Rate

15.9%

Rev. Share

Avg Spend by Tier (\$)

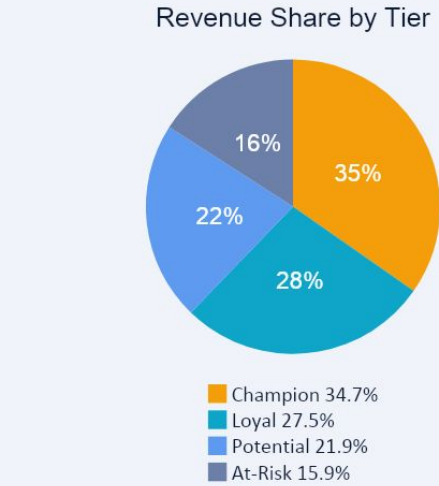
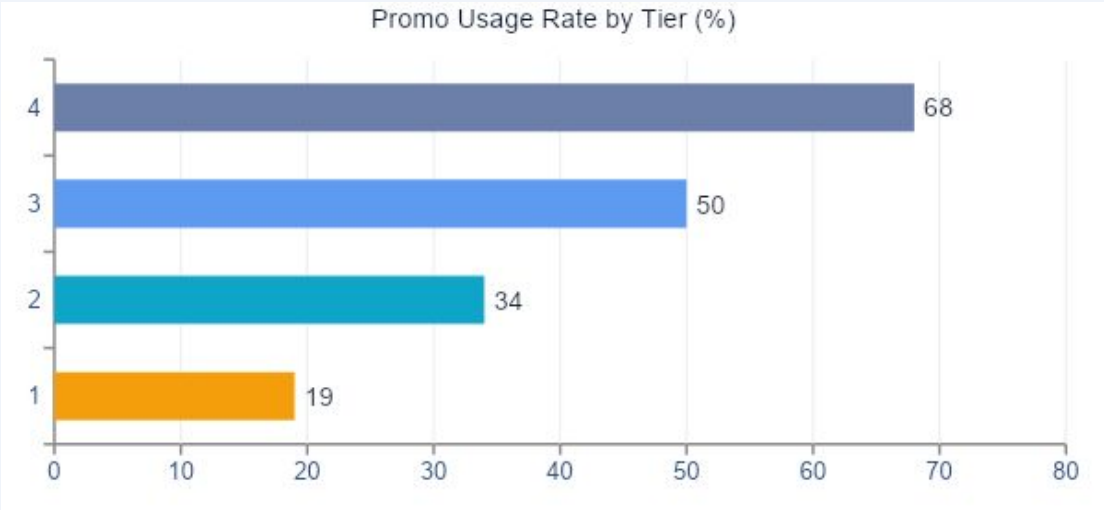


Key Insights

- Champions spend 2.2× more than At-Risk customers (\$82.91 vs \$37.90)
- Champions have only 19% promo rate — vs. 68% for At-Risk
- Top 50% of customers generate ~62% of total revenue
- True loyalty (high spend) is inversely related to promo-dependence
- Promo rate rises monotonically from Champion → At-Risk

Promotions: Friend or Foe?

43% of customers use discounts — yet Champions (top 25%) have only 19% promo rate. True loyalty and promo-dependence are inversely correlated.



Champion
19% promo

Reduce discount depth by 10%. Replace with loyalty points + early access. Minimal volume risk.

Loyal
34% promo

Shift cash discounts → value-adds (free shipping, gift-with-purchase). Protect margins.

Potential
50% promo

Behaviour-triggered coupons only (cart abandonment, 90-day inactivity). No blanket sales.

At-Risk
68% promo

Maintain targeted exit coupons. Phase last. High churn risk if discounts cut abruptly.



Margin Erosion

A 30% blanket discount on 45% margin products requires 3× volume to break even. Every blanket sale shrinks per-unit profit.



LTV Drag

Discount-acquired customers show lower lifetime value — they return for deals, not brand affinity, and lapse when sales end.



Price Anchor Effect

Continued blanket sales permanently lower customers' reference price, destroying the brand's future full-price pricing power.

Organic Demand vs. Discount-Driven Volume

Opportunity Score = z(Avg Spend) - z(Promo Rate) → High score = genuine brand pull at near-list price | Low score = discount-dependent demand

HIGH OPPORTUNITY — Invest Here

Arizona

Opp Score: 3.84

Spend: \$66.55

Promo: 34%

Prev Purch: 28.4

Alaska

Opp Score: 3.04

Spend: \$67.6

Promo: 40%

Prev Purch: 28.1

Virginia

Opp Score: 1.96

Spend: \$62.88

Promo: 38%

Prev Purch: 23.5

Pennsylvania

Opp Score: 1.93

Spend: \$66.57

Promo: 45%

Prev Purch: 27.4

Tennessee

Opp Score: 1.89

Spend: \$61.97

Promo: 36%

Prev Purch: 26

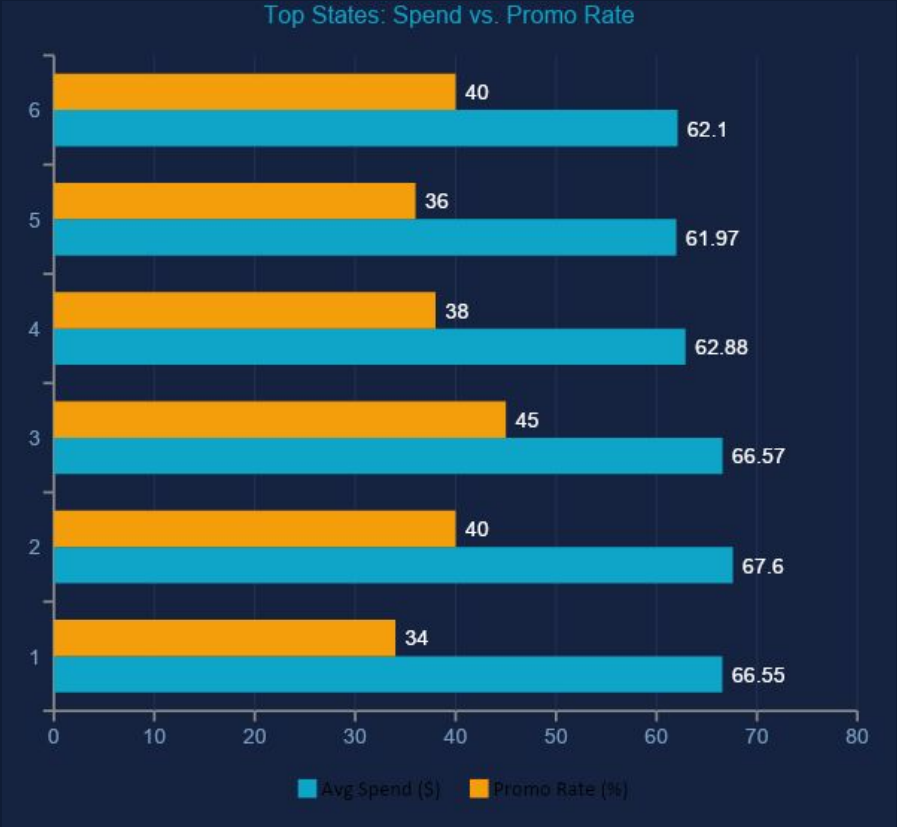
Michigan

Opp Score: 1.33

Spend: \$62.1

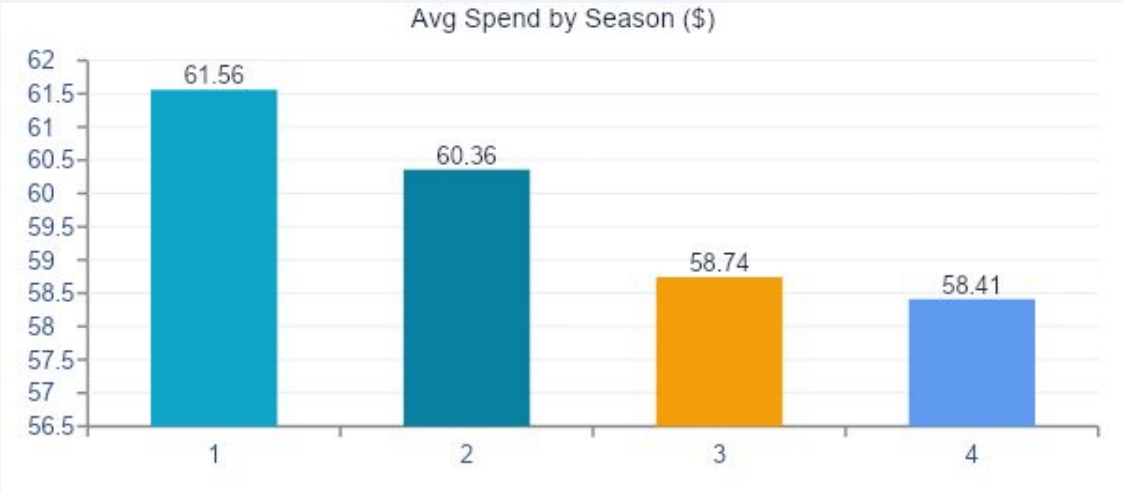
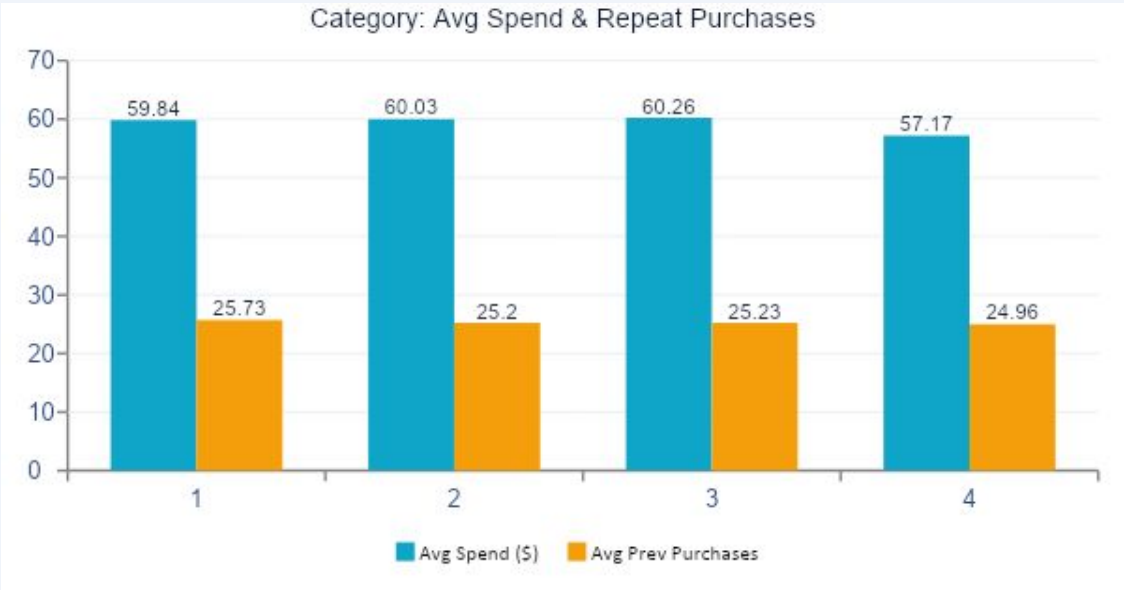
Promo: 40%

Prev Purch: 26.9



LOW OPPORTUNITY — Rethink Discount Mix

Category Funnel & Value Predictors



Accessories

Retention Anchor · Rev: 31.8%

Highest repeat purchase rate (25.73). Lower price point enables frequent buying — high-frequency retention driver. Bundle with outerwear to extend tenure.

Clothing

Core Volume Driver · Rev: 44.7%

44.7% of total revenue — the brand's commercial backbone. Strong across all segments. Key cross-sell and upsell opportunity for all tiers.

Footwear

Premium Signal · Rev: 15.5%

Highest avg spend (\$60.26). Typically purchased by committed customers. Strong affinity indicator — buyers tend to be multi-category.

Outerwear

Entry / Seasonal · Rev: 7.9%

Lowest avg spend (\$57.17) and repeat purchases (24.96). Likely seasonal/trial purchases. Weakest retention signal — pair with accessories bundles.

Strongest Value Predictors

01

Previous Purchases

[Very High]

Strongest predictor — repeat history predicts repeat behaviour

02

Purchase Amount

[High]

Direct revenue proxy — current spend predicts future spend

03

Purchase Frequency

[High]

Weekly/bi-weekly buyers over-represented in top value tier

04

Discount Independence

[Moderate]

Full-price buying signals sustainable willingness-to-pay

05

Review Rating

[Moderate]

Higher satisfaction correlates with repeat purchases + spend

06

Age / Gender / Payment

[Low]

Minimal predictive power — demographics don't segment value

Four-Panel Founder Dashboard — Blueprint

01

Customer Value Pyramid

Bar/funnel: 4 tiers from Champion → At-Risk. Revenue share per tier. Shows concentration instantly — top 25% = 34.7% of revenue.

KPIs: Tier size · Revenue share · AOV per tier · Subscription rate

Decision Use → Track tier migration over time; shift marketing budget toward Champion growth.

02

Promo Dependency vs. Loyalty Matrix

2×2 scatter: X = Promo Dependency Score, Y = Loyalty B Score. 4 quadrants: True Loyalists · Discounted Loyalists · Bargain Hunters · At-Risk.

KPIs: % customers per quadrant · Revenue per quadrant · Quadrant shift rate

Decision Use → Identify 'Discounted Loyalists' — move them to 'True Loyalists' by switching incentive type.

03

Geographic Opportunity Map

US state choropleth: colour = avg spend intensity, transparency = promo rate. AZ, AK, VA, MI highlighted as organic-demand markets.

KPIs: State AOV · Promo rate · Customer count · Opportunity score

Decision Use → Guide ad budget allocation: invest in organic-demand states; reduce promo depth elsewhere.

04

Category Retention Funnel

Horizontal funnel ranked by avg previous purchases. Entry categories (Outerwear) vs. retention anchors (Accessories) made visible.

KPIs: Avg prev. purchases · Revenue share · Promo rate · Cross-sell conversion

Decision Use → Merchandising decisions: bundle outerwear with accessories to extend customer tenure.

Phased Promotional Sunset — 3-Step Plan

Phase 1 Months 1–2	Champions Trigger: High Loyalty B score + Promo rate < 25% Reduce discount depth by 10% (e.g. 30% off → 20% off). Introduce double loyalty points and exclusive early access to new collections. No blanket coupon codes. Communicate the change as a 'VIP upgrade'.	Risk: Minor volume dip expected (<10%). Champions likely buy regardless of small discount cuts. If AOV drops > 8% in month 1, pause. Track: AOV · Repeat purchase rate · Margin per Champion +4–6% margin improvement
Phase 2 Months 3–5	Loyal Segment Trigger: Mid-high Loyalty B + Promo rate 30–45% Replace cash discounts with value-adds: free priority shipping, gift-with-purchase accessories, exclusive member sale access. Cash discounts only post-purchase for feedback collection. Promote subscription sign-up as the alternative hook.	Risk: Moderate risk — this segment shows ~34% promo sensitivity. If subscription sign-ups fall > 15%, add a loyalty points booster to compensate. Track: Subscription sign-ups · Revenue per Loyal customer · Shipping upsell rate +2–4% margin improvement
Phase 3 Months 6–12	Potential + At-Risk Trigger: Inactivity > 90 days OR cart abandonment event triggered Behaviour-triggered coupons only: 15% off on 90-day inactivity, 24hr cart abandonment follow-up offer. No sitewide sales. Gradually extend inactivity window: 90 → 120 → 150 days over 6 months to wean off blanket discount expectation.	Risk: Highest risk: price-sensitive segments. Accept a 15–20% purchase frequency decline. Goal is improved margin quality per remaining purchase, not volume preservation. Track: Triggered offer redemption rate · Purchase frequency trend · Promo code cost per order Margin quality ↑; volume dip accepted

Critical: A/B test discount depth within each segment before full rollout. Pause further cuts if revenue loss exceeds phase threshold.

The Brand's Best Customer — A Data-Backed Portrait



Demographics

Age 35–55 · Avg age 44 · Equal gender split



Geography

Arizona · Alaska · Virginia · Michigan



Spend Profile

Avg \$82.91/order · 35+ previous purchases



Satisfaction

Review rating 3.80+ · Positive brand sentiment



Champion

Ideal Customer



Purchase Behaviour

Weekly / bi-weekly frequency · Only ~19% use discounts · Multi-category buyer



Category Mix

Accessories + Clothing primary · 25+ lifetime items · Cross-category purchaser



Loyalty Signals

Subscription-eligible · Responds to exclusivity & convenience, not price cuts



Acquisition Path

Lookalike targeting in AZ / MI · Brand-story content · Loyalty club as hook

How to Acquire More Champions:



Run lookalike ads targeting AZ / MI / VA profiles



Lead with loyalty club hook ('Free shipping, earn points') — not a discount offer



Influence via brand-story content, not coupon-led creative



A/B test subscription as new-customer acquisition funnel

From Discount-Dependent to Loyalty-Driven Growth

The business has a genuine loyal core — but its loyalty has been propped up by discounts. The path forward is phased, targeted, and segment-specific: protect Champions, convert Loyals, trigger Potentials.

R1

 **Adopt Commercial Loyalty (Definition B)**

Use Loyalty B as the primary lens for all marketing, segmentation, and CRM. Definition A may supplement as an engagement monitor only.

Trade-off: Requires re-instrumenting CRM and re-tagging existing segments.

R2

 **Execute Phased Promo Sunset**

3-phase reduction over 6–12 months. Champions first → Loyal → Potential/At-Risk. Use behaviour-triggered offers, not ongoing campaigns.

Trade-off: Short-term volume dip in Phases 2–3. Plan revenue bridge through margin expansion.

R3

 **Invest in High-Opportunity Geographies**

Double ad spend in AZ, MI, VA, TN. Use Champion ICP lookalike targeting with brand-story creative, not discount-led campaigns.

Trade-off: Higher upfront CPM in premium states. Offset by higher LTV from acquired customers.

R4

 **Build Subscription as Loyalty Backbone**

Subscription rate is only 27%. Grow it by offering free shipping + loyalty points as new-customer acquisition hook. Target Potential/Loyal for upgrade.

Trade-off: Subscription perks have upfront cost. Model break-even at 2nd purchase for CAC recovery.

R5

 **Launch Accessories as Retention Category**

Accessories have the highest repeat purchase rate (25.73). Bundle with Outerwear (lowest retention, 24.96) to extend category engagement and accelerate customer tenure. Cross-sell is the organic growth lever available immediately.

Trade-off: Bundle discounts may reduce per-unit margin. Net LTV gain likely offsets — A/B test before full rollout.

5 Business Questions → 5 Traceable Queries

Q1

Who are genuinely loyal vs. discount-only customers?

Approach: CTE-based loyalty B scoring. Filter: loyalty_b_tier = 'Champion' AND promo_rate < 0.25 AND prev_purchases > 30. Cross-tab with discount usage to separate true loyalists.

Finding:

Champions: 975 customers · \$82.91 avg spend · 19% promo rate · 35.15 avg prev purchases

Q2

Which behavioural patterns predict high customer value over time?

Approach: Pearson correlation via COVAR_SAMP / STDDEV_SAMP. Join on feature-engineered table. Compute correlations of all 18 variables with Purchase Amount.

Finding:

Previous Purchases (r=0.53) and Purchase Amount itself (r=0.72) are top predictors. Demographics add no signal.

Q3

Which geographies are commercially underlevered?

Approach: GROUP BY Location. Compute z-score of avg_spend and promo_rate per state. Opportunity score = z(spend) – z(promo). Rank DESC. Filter top decile.

Finding:

AZ (3.84) · AK (3.04) · VA (1.96) · PA (1.93) · TN (1.89) are top 5 opportunity states.

Q4

How should the brand restructure its promotional strategy?

Approach: Segment × promo cross-tab. CASE WHEN loyalty_b_tier = 'Champion' THEN 'Reduce 10%' WHEN tier='Loyal' THEN 'Value-adds' WHEN tier='Potential' THEN 'Triggered only' ELSE 'Exit coupons' END.

Finding:

Champions: cut depth. Loyal: value-adds. Potential: triggered. At-Risk: exit coupons only. No blanket sitewide sales.

Q5

What does the ideal customer profile look like for acquisition targeting?

Approach: WHERE loyalty_b_tier = 'Champion': SELECT AGG(Age), MODE(Location), MODE(Category), AVG(Previous_Purchases), AVG(Purchase_Amount), SUM(Discount)/COUNT(*) as promo_rate, AVG(Review_Rating).

Finding:

Age 35–55 · AZ/AK/VA/MI · Accessories+Clothing · 35+ prev purchases · \$82.91 AOV · 19% promo rate · 3.80 rating

Key Findings & Recommendations at a Glance

Central Finding: The business is building loyalty — but with discount crutches. Champions are real, repeat, full-price buyers. But the brand has also trained a large discount-dependent segment that responds to deals, not brand affinity. The path forward: protect Champions, wean Loyals off discounts, trigger Potentials selectively.



Definition B Selected

- $r=0.723$ correlation with spend vs $r=0.004$ for Definition A
- Champions: \$82.91 avg spend, 19% promo rate, 35+ prev purchases
- At-Risk: \$37.90 avg spend, 68% promo rate, 15.6 prev purchases
- Promo rate declines monotonically Champion → At-Risk



Geographic Opportunity

- Arizona (opp: 3.84) — \$66.55 avg spend, only 34% promo rate
- Alaska (opp: 3.04) — \$67.60 avg spend, 40% promo rate
- Virginia, Pennsylvania, Tennessee round out top 5
- Indiana, Oregon: discount-dependent, below-avg spend — de-prioritise



Category Intelligence

- Accessories: highest repeat rate (25.73) — retention anchor
- Clothing: 44.7% of revenue — core volume driver
- Outerwear: lowest spend (\$57.17) & repeat (24.96) — entry/seasonal
- Fall generates highest avg spend (\$61.56) — prime campaign season

5 Strategic Recommendations

R1	Adopt Definition B	Use commercial loyalty as primary segmentation lens
R2	Execute Phased Promo Sunset	3-phase reduction: Champions (M1-2) → Loyal (M3-5) → Potential/At-Risk (M6-12)
R3	Invest in Organic Geo Markets	Double spend in AZ, MI, VA — lookalike targeting, brand-story creative
R4	Grow Subscription Base	Use subscription as acquisition hook; target Potential/Loyal for upgrade
R5	Accessories × Outerwear Bundle	Bundle high-retention (accessories) with low-retention (outerwear) to extend tenure



From Data to Decisions

Decoding Customer Value · A SQL-Driven Retention Strategy



Def B Selected

$r = 0.723$
vs spend



4 Segments

Champion → At-Risk
value pyramid



6 Geo Markets

AZ · AK · VA · MI
organic demand



3-Phase Plan

6–12 month
phased sunset



ICP Defined

Age 35–55
full-price buyer